A finite-witness architecture for ordinary Collatz

Countable certificate search and conditional limit exchange

Collatz-Final project
github.com/randogelist/Collatz-Final

Abstract

We document a sufficient finite-certificate route to the ordinary Collatz conjecture. A terminating checker tests natural affine matrix interpretations of a seven-symbol mixed binary–ternary rewriting system. The repository proves in Lean that acceptance implies Collatz, serializes every accepted certificate into a natural-number search space, and proves that symbolic pruning preserves acceptance. The required project architecture is the forward implication from accepted-witness existence; a converse is not required. No accepted global certificate is supplied. We also describe Frontier A's separate dyadic predecessor formulation: two explicitly identified iterated limits commute exactly when Collatz holds. That interchange remains an open premise, and its supporting source branch is outside the present repository CI target.

1 Main result and scope

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$. Let

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ 3n+1, & n \equiv 1 \pmod 2, \end{cases} \qquad \mathsf{Collatz}: \ \forall n \geq 1, \ \exists t \in \mathbb{N}, \ C^t(n) = 1.$$

For each actual dimension $m \geq 1$, let $\mathcal{W}_m$ be the finite-table certificate type defined in Section 2. The Boolean function check is total and consists entirely of finitely many exact natural-number comparisons.

Theorem 1.1 (Finite-witness soundness)**.** *The checked source establishes*

$$\boxed{(\exists m \geq 1, \ \exists W \in \mathcal{W}_m, \ \mathsf{check}(W) = \text{true}) \implies \mathsf{Collatz}.}$$

Lean declarations: `CHECKER_SOUND`; `FINITE_WITNESS_IMPLIES_COLLATZ`.

The implication contains no predecessor-density, analytic regularity, trajectory-coverage, or termination premise. Its universal arithmetic transfer is proved downstream of the finite check. Section 3 gives an exhaustive natural-number encoding of this certificate language, and Section 4 removes entire impossible families by symbolic proof.

The mathematical goal left by this architecture is concrete: exhibit one finite table that passes the checker, and certify its admission in Lean. Soundness does not itself establish that such a table exists. This preprint reports a conditional proof architecture, not a proof of the Collatz conjecture.

2 The finite matrix certificate

The alphabet is

$$\Sigma = \{L, R, f, t, 0, 1, 2\}.$$

Its mixed binary–ternary encoding is taken from Yolcu, Aaronson, and Heule [1]. The following are the eleven oriented rules used by the repository:

Role	Rules
Dynamics	$fR \to R, \quad tR \to 2R$
Digit transport	$f0 \to 0f, \quad f1 \to 0t, \quad f2 \to 1f$
Digit transport	$t0 \to 1t, \quad t1 \to 2f, \quad t2 \to 2t$
Left boundary	$L0 \to Lt, \quad L1 \to Lff, \quad L2 \to Lft$

A certificate $W \in \mathcal{W}_m$ assigns to every symbol s a matrix $M_s \in \mathbb{N}^{m\times m}$ and a vector $b_s \in \mathbb{N}^m$. Write $F_s(x) = M_s x + b_s$. For a word $w = s_1 \cdots s_\ell$, put $F_w = F_{s_1} \circ \cdots \circ F_{s_\ell}$; thus the leftmost symbol acts last. The empty word is the identity, and compilation uses

$$(M_A, b_A) \circ (M_B, b_B) = (M_A M_B,\ M_A b_B + b_A).$$

The checker accepts precisely when

$$(M_s)_{00} \geq 1 \qquad (s \in \Sigma), \tag{1}$$

$$M_{\mathrm{rhs}(r)} \leq M_{\mathrm{lhs}(r)}, \qquad b_{\mathrm{rhs}(r)} \leq b_{\mathrm{lhs}(r)} \qquad (r \text{ any rule}), \tag{2}$$

$$(b_{2R})_0 < (b_{tR})_0. \tag{3}$$

In (2), both comparisons are entrywise. There are $7m^2+7m$ raw coefficients and $7+11(m^2+m)+1$ scalar comparisons. The propositional interface `Check W` is not a field of the raw witness.

Lean declarations: `check_iff`; `toRanking`.

Proof outline of Theorem 1.1. Define $\rho(w) = (F_w(0))_0 \in \mathbb{N}$. Nonnegative coefficients give monotonicity in every coordinate; (1) preserves strict decrease in coordinate zero through every prefix. Hence every rule weakly decreases ρ in every prefix and suffix context, while $tR \to 2R$ strictly decreases it.

The source constructs binary numerals: L represents 1, and appending f or t represents $2n$ or $2n+1$. The nine auxiliary rules normalize $3n+j$, $j \in \{0,1,2\}$, preserving the required weak rank inequality. For the shortcut map $T(2n) = n$, $T(2n+1) = 3n+2$, even steps are weakly decreasing and nonbase odd steps are strictly decreasing. Strong induction on rank, with structural induction on binary numerals for the even steps, proves that every positive numeral reaches 1. Binary coverage handles every positive integer. Finally, each T step is one or two ordinary C steps, giving the stated conclusion. □

Lean declarations: `ranking_implies_collatz`; `shortcut_implies_full`.

3 An exhaustive countable search

The source convention `Witness d` means actual dimension $d+1$. After excluding scalars, the executable raw format is simply

$$\mathsf{RawWitness} = \mathbb{N}^3, \qquad R = (q, a, b).$$

Here q selects actual dimension $q+2$, while a and b encode all seven matrices and all seven bias vectors. No proposition, proof object, oracle, or infinite table is stored in R.

Use the pairing map

$$\pi(u, v) = \frac{(u+v)(u+v+1)}{2} + u, \qquad \mathrm{code}(q, a, b) = \pi(q, \pi(a, b)).$$

The implementation defines the triangular number recursively and constructs a diagonal decoder `unpair`. It proves $\mathrm{unpair}(\pi(u, v)) = (u, v)$. Recursive pairing therefore encodes any finite table, with an exact decoding theorem.

The decoder fills each mandatory entry $(M_s)_{00}$ with its stored value plus one. Packing an accepted certificate subtracts one from these seven entries; positivity makes the reconstruction exact. Let $\mathrm{candidate} : \mathbb{N} \to \mathsf{RawWitness}$ be the nested decoder. Then

$$\mathrm{candidate}(\mathrm{code}(R)) = R.$$

This proves coverage of all raw candidates. Together with scalar exclusion and the coupling guard of Section 4, it also proves coverage of every accepted original certificate.

Theorem 3.1 (Acceptance-preserving enumeration)**.** *For the total executable checker* $\mathsf{checkCode}$*,*

$$\boxed{(\exists n \in \mathbb{N},\ \mathsf{checkCode}(n) = \mathrm{true}) \iff (\exists m \geq 1,\ \exists W \in \mathcal{W}_m,\ \mathsf{check}(W) = \mathrm{true}).}$$

Consequently, $(\exists n,\ \mathsf{checkCode}(n) = \mathrm{true}) \Longrightarrow \mathsf{Collatz}$*.*

Lean declarations: `INDEXED_SEARCH_IFF_ORIGINAL_ACCEPTANCE`; `INDEXED_SEARCH_IMPLIES_COLLATZ`.

Proof. An accepted index decodes to an accepted matrix table. Conversely, every accepted table is nonscalar, packs into the raw format, and is recovered by decoding. The guard rejects no accepted table. Apply Theorem 1.1. □

The bounded routine checks the prefix below its bound, starting from the largest index:

$$\mathsf{findAccepted}(0) = \mathrm{none}, \qquad \mathsf{findAccepted}(b+1) = \begin{cases} \mathrm{some}(b), & \mathsf{checkCode}(b) = \mathrm{true}, \\ \mathsf{findAccepted}(b), & \text{otherwise.} \end{cases}$$

The source proves both

$$\begin{aligned} \mathsf{findAccepted}(b) = \mathrm{some}(n) &\implies n < b \ \wedge\ \mathsf{checkCode}(n) = \mathrm{true}, \\ \mathsf{checkCode}(n) = \mathrm{true} &\implies \mathsf{findAccepted}(n+1) = \mathrm{some}(n). \end{aligned}$$

Thus a finite search succeeds exactly when this certificate language has an accepted witness. Exhaustiveness is language-relative; it is not a guarantee of success. The reference decoder is intentionally transparent and may be impractical on large codes. No search-performance claim is made.

4 Symbolic reductions of the witness space

The exclusions below are proved for all coefficient values. They do not result from testing a bounded list of candidates.

Proposition 4.1 (Scalar obstruction)**.** *No one-dimensional natural affine interpretation passes the checker.*

Proof. Let a, b, f, t, u be the slopes for $L, R, f, t, 2$, respectively. Write β_s for the bias of symbol s. The odd rule and the $L2$ rule imply

$$\begin{aligned} ub &\le tb, & aft &\le au, \\ a(f\beta_t + \beta_f) + \beta_L &\le a\beta_2 + \beta_L, \\ u\beta_R + \beta_2 &< t\beta_R + \beta_t. \end{aligned}$$

Since $a, b, f \ge 1$, cancellation and nonnegativity give

$$u \le t \le ft \le u.$$

Therefore $u = t$. The weak bias inequality gives $\beta_t \le f\beta_t + \beta_f \le \beta_2$, whereas the strict odd inequality gives $\beta_2 < \beta_t$. This is a contradiction. □

Lean declarations: `SCALAR_CHECKER_REJECTS`; `ACCEPTED_DIMENSION_POSITIVE`.

Proposition 4.2 (Necessary coupling)**.** *Every accepted certificate has*

$$\exists s \in \{L, t, 2\},\ \exists j > 0 :\ (M_s)_{0j} > 0.$$

Proof. Suppose instead that the first rows of M_L, M_t, M_2 have no off-diagonal entries. Their first coordinates then behave as scalar affine maps. The remaining contributions from M_f and b_f are nonnegative and may be discarded to obtain the scalar inequalities used above. The same contradiction follows. No isolation condition is imposed on M_f, M_R, or the other rows. □

Lean declarations: `THREE_SCALAR_ROWS_IMPOSSIBLE`; `ACCEPTED_REQUIRES_FIRST_ROW_COUPLING`.

In particular, every fully diagonal interpretation is excluded. Let $G(W)$ be the finite Boolean test of the displayed coupling condition. The reduced checker satisfies the exact identity

$$G(W)\ \&\&\ \mathsf{check}(W) = \mathsf{check}(W).$$

The raw search therefore starts at dimension two, enforces the seven positive leading entries by construction, and applies G before the full checker. These reductions preserve all accepted original certificates; no claim of optimality or global minimality is made.

The remaining finite problem

For some dimension $m \ge 2$, find seven natural matrices and seven natural vectors satisfying (1)–(3). This one accepted table would suffice for ordinary Collatz by the proved transfer. A search engine may propose it, but admission must be checked for the exact data and connected to the audited theorem. At the source snapshot documented here, no such global witness has been supplied.

5 Frontier A: conditional limit-exchange closure

This section concerns the separate supporting archive [3], not the eight-module CI target. The construction uses the actual ordinary map C.

Let $B = \{n \geq 1 : \forall t,\ C^t(n) \neq 1\}$. For $n > 1$, define

$$\tau(n) = \min\{t : (C^t(n))^2 \leq n\}$$

when that set is nonempty. Put $J(n) = C^{\tau(n)}(n)$ in that case, and $J(n) = 0$ otherwise; set $J(0) = 0$, $J(1) = 1$. Zero marks genuine failure of a crossing, not a finite computational timeout. For successful restarts of $n > 1$, the arithmetic proof gives

$$0 < J(n), \qquad J(n)^2 \leq n < 4J(n)^2, \qquad J(n) < n.$$

Define

$$A_k = \{n \geq 1 : J^k(n) = 0\}, \qquad U_c(j,k) = \mathbf{1}_{A_k}(2^j c) \quad (c \geq 1).$$

The sets A_k increase, and $\bigcup_k A_k = B$: successful restarts of a bad integer remain bad and strictly decrease, so they cannot continue forever; good integers cannot encounter a failed restart.

Theorem 5.1 (Both limit values)**.** *For every positive c,*

$$\lim_{j\to\infty} \lim_{k\to\infty} U_c(j,k) = \mathbf{1}_B(c), \qquad \lim_{k\to\infty} \lim_{j\to\infty} U_c(j,k) = 0.$$

Proof. The identity $C^j(2^j c) = c$ makes $2^j c$ bad exactly when c is bad. For each fixed j, monotone absorption therefore eventually equals $\mathbf{1}_B(c)$ as k increases.

For $P_c = \{n \geq 1 : \exists t,\ C^t(n) = c\}$, the supporting source proves

$$n \in P_c \cap A_k \quad \Longrightarrow \quad n < (2c)^{2^k}.$$

Thus $U_c(j,k) = 1$ requires $2^j c < (2c)^{2^k}$. For each fixed k, this fails at all sufficiently large j. One explicit sufficient cutoff in the source is $j \geq (2c)^{2^k}$. The two outer limits are consequently constant. □

Lean declarations: `DEPTH_THEN_TOWER_LIMIT`; `TOWER_THEN_DEPTH_LIMIT` in `CollatzNativeLimits`.

Theorem 5.2 (Conditional closure)**.** *Equality of these two iterated limits for every $c \geq 1$ is equivalent to* Collatz*.*

Proof. Equality forces $\mathbf{1}_B(c) = 0$ for every positive c. Conversely, Collatz makes the entire array zero. □

The interchange premise remains unproved. Lean represents these limits by exact eventual equality of natural-valued sequences; the ordinary real-limit interpretation follows for values $0, 1$. The restart and absorption predicates are noncomputable. This tower theorem has no predecessor-density input. The original harmonic Frontier A instead retains that separate input; these two formulations must not be conflated.

6 Verification record and reproducibility

The main certificate chain was checked by GitHub Actions using Lean 4.33.0. The pinned compiler and source snapshot for this preprint are:

```
Compiler:  d8b18978322de05a8f3dba51ef03cf5461676c17
Source:    f317167a55f7c6b5a815837f5a84a3c693946744
```

Run 40175415091 completed successfully: eight modules compiled, 24 endpoints passed the transitive axiom audit, and four audit-guard tests passed. The allowed foundational dependencies are `propext`, `Classical.choice`, and `Quot.sound`. No research axiom or `sorryAx` is allowed by the audit.

Claim	Evidence scope
Accepted finite witness implies ordinary Collatz	Current repository CI
Indexed acceptance iff matrix-certificate acceptance	Current repository CI
Scalar exclusion and coupling guard	Current repository CI
Bounded-search soundness and coverage	Current repository CI
Frontier A limit identities and conditional equivalence	Supporting archive; outside current CI
An accepted global certificate	Not supplied

The eight modules are:

```
Definitions, FiniteAlgebra, MatrixWitness, Dynamics,
CrownMain, MatrixObstruction, RawSearch, SearchMain.
```

The workflow installs a checksum-pinned Lean release, validates compatible cache entries, performs a fresh endpoint audit, and retains receipts. Assistant-run Lean checks use CI only.

Observed runner output

The following lines are from the successful run above, with their common version prefix omitted:

```
ACCEPTED_WITNESS_EXISTS_IMPLIES_COLLATZ=PROVED
COLLATZ_IMPLIES_ACCEPTED_WITNESS_EXISTS=NOT_PROVED
ACCEPTED_WITNESS_EXISTS_IFF_COLLATZ=NOT_ESTABLISHED
ACCEPTED_CONCRETE_WITNESS=NOT_SUPPLIED
```

The project requires only the proved forward implication. Its remaining task is an accepted global witness.

References

[1] E. Yolcu, S. Aaronson, and M. J. H. Heule. *An Automated Approach to the Collatz Conjecture*. arXiv:2105.14697, 2021; version 3, 2022. arxiv.org/abs/2105.14697. Source of the mixed binary–ternary encoding.

[2] Collatz-Final project. *Finite-witness countable search: Lean sources and CI receipts*. Repository snapshot f317167a55f7, 25 October 2026. Exact theorem names and source provenance appear in the repository.

[3] Collatz-Final project. *Frontier A interchange*, supporting source archive, 17 October 2026. `COLLATZ_FRONTIER_A_INTERCHANGE_LEAN_v0_1_0.zip`. The archive records historical Linux compilation; it does not certify a current GitHub run. SHA-256: `096e6a67666adc00cbe3fdac80d162a1b2d7a8e9670411fe8e0d2b38490833ae`.